

## The $\mathfrak{A}^\dagger \bar{\mathfrak{C}}^\dagger \times \mathfrak{B}_2$ row

Transverse divergences, the  $p^+$  integral, and whether the  $+$  prescription may be used

$$S \equiv p^+ + k^+, \quad A \equiv (x - z)^2, \quad B \equiv (x - w)^2, \quad C \equiv (z - w)^2, \quad \beta \equiv \frac{k^+ B}{A}, \quad \kappa \equiv \frac{k \cdot (y' - z)}{k^+}$$

**Verdict.** The prefactor, the  $\mathfrak{C}$  delta, the phase split and the tensor contraction are all **correct except one term**: the second half of your  $\frac{1}{p^+}$  bracket carries the wrong vector (§2). On divergences: **no ultraviolet divergence** anywhere — all 26 collapsing regions converge (§3) — and, unlike the previous rows, **no infrared one either**: the  $z$  integral converges as it stands, because the phase  $e^{-ik \frac{p^+}{k^+} (y' - z)}$  damps it (§4). The  $p^+$  integral is clean, six exponential–integral masters, no pole on the contour (§5). **But the answer to your last question is no.** The  $+$  prescription is precisely what must *not* be applied naively here: the  $p^+ \rightarrow 0$  subtraction deletes that phase and **creates** an infrared divergence the row does not have (§6).

### 1. Prefactor, delta and phase — all exact

| ingredient      | value                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| phases          | $(-i)_{\mathfrak{A}^\dagger} (-1)_{\bar{\mathfrak{C}}^\dagger} (+i)_{\mathfrak{B}_2} = -1 \rightarrow$ the overall minus $\checkmark$ |
| numeric         | $\frac{4}{(2\pi)^3} \frac{1}{\sqrt{2\pi}} \frac{1}{2\pi\sqrt{2}} \frac{1}{4\pi^2} = \frac{1}{(2\pi)^3} \frac{1}{4\pi^4} \checkmark$   |
| momentum        | $\frac{1}{\sqrt{S}} \cdot \frac{1}{\sqrt{S}} \frac{\sqrt{p^+ k^+}}{S} \cdot \sqrt{p^+ k^+} = \frac{p^+ k^+}{S^2}$                     |
| after the delta | kernel $\frac{k^+}{S}$ , Jacobian $\left(\frac{S}{k^+}\right)^2 \rightarrow$ net $\frac{p^+}{S} \checkmark$                           |

The  $\mathfrak{C}$  delta  $\delta^{(2)}[(y' - w') + \frac{p^+}{S}(w' - z)]$  gives  $w' = \frac{Sy' - p^+ z}{k^+}$  and  $w' - z = \frac{S}{k^+}(y' - z)$ , hence

$$w' - w = (y' - w) + \frac{p^+}{k^+}(y' - z) \implies e^{-ik \cdot (w' - w)} = e^{-ik \cdot (y' - w)} e^{-ik \cdot \frac{p^+}{k^+} (y' - z)} \quad \checkmark$$

exactly your two–factor phase. **Remember that second factor** — it is the whole story of §4 and §6.

### 2. The contraction — one term is wrong

With  $M^{ij} \equiv \frac{\delta^{ij}(A-B)}{2SC} + \frac{\mathcal{K}^{ij}}{k^+} + \frac{\mathcal{P}^{ij}}{p^+}$ ,

$$\mathcal{K}^{ij} = \frac{(x - z)^j (z - w)^i}{C} - \frac{(x - z)^j (x - w)^i}{2B}, \quad \mathcal{P}^{ij} = \frac{(x - w)^i (z - w)^j}{C} + \frac{(x - w)^i (x - z)^j}{2A},$$

contracting the  $\mathfrak{C}$  bracket  $[\delta_{k'm} \delta_{ji} - \frac{S}{p^+} \delta_{jm} \delta_{k'i} - \frac{S}{k^+} \delta_{im} \delta_{k'j}]$  over  $i, j$  and multiplying by the  $\frac{p^+}{S}$  of §1 gives

$$\frac{p^+}{S} \delta_{k'm} \text{tr } M - M^{k'm} - \frac{p^+}{k^+} M^{mk'}.$$

Expanded, this is your six–term bracket. Five terms match exactly, including the two  $-\frac{(A-B)}{2C}$  pieces inside your  $\frac{p^+ \delta_{k'm}}{k^+ S}$  and  $\frac{\delta_{k'm}}{S}$  brackets, which come from the  $\delta^{ij}$  part of  $M$  feeding into the  $-\frac{S}{p^+}$  and  $-\frac{S}{k^+}$  structures of  $\mathfrak{C}$ . The exception:

$$\begin{aligned} \text{you wrote} & -\frac{1}{p^+} \left[ \frac{(x - w)^{k'} (z - w)^m}{(z - w)^2} + \frac{(x - z)^m (z - w)^{k'}}{2(x - z)^2} \right] \\ \text{it should be} & -\frac{1}{p^+} \left[ \frac{(x - w)^{k'} (z - w)^m}{(z - w)^2} + \frac{(x - z)^m (x - w)^{k'}}{2(x - z)^2} \right] \end{aligned}$$

because this term is  $-\frac{\mathcal{P}^{k'm}}{p^+}$ , and **both** halves of  $\mathcal{P}^{ij} = (x - w)^i [\frac{(z - w)^j}{C} + \frac{(x - z)^j}{2A}]$  carry the same  $(x - w)^i$ . The error is exactly

$$\Delta = +\frac{1}{p^+} \frac{(x - z)^{k'} (x - z)^m}{2(x - z)^2}, \quad \|\Delta\| = \frac{1}{2p^+} \text{ (configuration independent),}$$

which is why the measured discrepancy was the same number,  $1.3514 = \frac{1}{2 \times 0.37}$ , at every test configuration. With the fix the contraction matches to  $3 \times 10^{-16}$ .

2.1 Two smaller fixes

- **The measure.** You write  $\int_{w,w',y',z,y,x}$ . But  $w'$  has been eaten by the delta,  $y$  does not appear in the integrand, and  $x'$  is **missing** (it is in both  $\frac{(x'-y')^{k^+}}{(x'-y')^2}$  and  $\rho^{e'}(x')$ ). It should be  $\int_{x,x',y',z,w}$ .
- **Transposes.** The bra bracket is the one from the two- $\mathfrak{C}$  row and carries the same  $U^{d'b}(z) / U^{bd'}(z)$  and  $U^{e'c'}(y') / U^{c'e'}(x')$  mismatch. The ket bracket  $[f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)]$  is right, with  $f$  correctly *not* factored out (the two carry  $f^{cbd}$  and  $f^{acd}$ ).

3. No ultraviolet divergence

All 26 subsets of  $\{x, x', y', z, w\}$  collapsed at scale  $\rho$  against  $\rho^{2(n-1)} \frac{d\rho}{\rho}$ : **every one converges**. The region you had in mind,  $x, z, w \rightarrow$  a point where  $D = p^+ A + k^+ B$  vanishes like  $\rho^2$ , gives integrand  $\rho^{-2}$  against a three-point measure  $\rho^3 d\rho$ .

**Why the loop has no transverse UV.** The loop gluon carries  $p^+$  and is emitted at  $z$  in the ket ( $\mathfrak{B}_2$ ) and absorbed at  $z$  in the bra — the  $\mathfrak{C}$  delta puts the bra's emission point  $w'$  at  $\frac{Sy'-p^+z}{k^+}$ , so the loop closes at **the same transverse point on both sides**. There is no separation to shrink, hence no short-distance singularity. The loop's large-momentum behaviour is carried entirely by the  $p^+$  integration instead (§5), where the  $h = p^+$  master grows like  $\vee$ .

4. And no infrared one either — the phase protects  $z$

| variable | $R = 10^2$    | $10^3$        | $10^4$        | verdict                                                         |
|----------|---------------|---------------|---------------|-----------------------------------------------------------------|
| $ x $    | 0.287047      | 0.287065      | 0.287066      | log divergent — but $x$ is a $\rho$ position, cut by the source |
| $ x' $   |               |               | noise         | convergent                                                      |
| $ y' $   | 6.2e-2        | 1.0e-2        | 1.5e-3        | convergent                                                      |
| $ z $    | <b>6.1e-2</b> | <b>1.6e-2</b> | <b>5.4e-3</b> | <b>convergent</b>                                               |
| $ w $    |               |               | oscillatory   | conditionally convergent (it is the Fourier transform to $k$ )  |

**This row is different from the last three.** There,  $z$  carried no phase and the large- $|z|$  end gave a genuine infrared logarithm. Here the  $\mathfrak{C}$  delta leaves the factor  $e^{-ik \cdot \frac{p^+}{k^+}(y'-z)}$ , whose exponent grows with  $|z|$ , so the  $z$  integral is damped for every  $p^+ \neq 0$  and converges on its own. **Nothing needs regulating.**

**A note on  $|w|$ , stated carefully.**  $\mathcal{P}^{ij}$  grows like  $|w|$  through  $\frac{(x-w)^i(x-z)^j}{2A}$ , so the integrand falls only as  $\frac{1}{|w|}$ . The leading growth carries a single  $\hat{w}$ , which the phase  $e^{ik \cdot w}$  converts into a Bessel  $J_1$ -type term, so the radial integrand  $R \langle F \rangle$  has an  $R^{-1/2}$  envelope times  $\cos(|k|R - \text{phase})$ . Measured:

| $R$                     | 100    | 300    | 1000   | 3000   |
|-------------------------|--------|--------|--------|--------|
| $R  \langle F \rangle $ | 0.5474 | 0.2394 | 0.1669 | 0.0939 |

consistent with  $R^{-1/2}$  (the scatter is the oscillation). An oscillating  $R^{-1/2}$  radial integrand is **conditionally** convergent, Fresnel-like — and that is what the  $w$  integral should be, since it is the Fourier transform to the measured momentum ( $\int d^2w \frac{e^{ik \cdot w}}{|w|} = \frac{2\pi}{|k|}$ ). **Caveat:** this rests on the envelope measurement plus the analytic argument. A direct cumulative quadrature does *not* confirm it here — resolving the radial oscillation out to  $R \sim 10^3$  needs a radial step well below  $2\pi/|k| \approx 7.8$ , which the runs above do not have, so the partial sums wander. Do not bound this tail absolutely, and do not read a divergence into it either.

5. The  $p^+$  integral

Writing  $D = A(p^+ + \beta)$  with  $\beta = \frac{k^+ B}{A}$ , and noting  $A, B > 0$  so  $\beta > 0$  and **there is no pole on the contour**, the  $p^+$  dependence reduces to six masters. Because of the phase they are exponential integrals, not logarithms:

$$\mathcal{M}[h] = \int_{\Lambda}^{\vee-k^+} dp^+ \frac{h(p^+) e^{-i\kappa p^+}}{p^+ A + k^+ B}, \quad h \in \left\{ \frac{p^+}{s^2}, \frac{p^+}{s}, \frac{1}{s}, 1, \frac{1}{p^+}, p^+ \right\},$$

$$\int \frac{e^{-i\kappa p^+}}{p^+ + a} dp^+ = e^{i\kappa a} \text{Ei}(-i\kappa(p^+ + a)),$$

$$\mathcal{M}[1] = \frac{e^{i\kappa\beta}}{A} \text{Ei}(-i\kappa(p^+ + \beta)), \quad \mathcal{M}\left[\frac{1}{p^+}\right] = \frac{1}{A\beta} \left[ \text{Ei}(-i\kappa p^+) - e^{i\kappa\beta} \text{Ei}(-i\kappa(p^+ + \beta)) \right],$$

$$\mathcal{M}[p^+] = \frac{1}{A} \left[ \frac{e^{-i\kappa p^+}}{-i\kappa} - \beta e^{i\kappa\beta} \text{Ei}(-i\kappa(p^+ + \beta)) \right], \quad \mathcal{M}\left[\frac{1}{S}\right] = \frac{e^{i\kappa k^+} \text{Ei}(-i\kappa(p^+ + k^+)) - e^{i\kappa\beta} \text{Ei}(-i\kappa(p^+ + \beta))}{A(\beta - k^+)},$$

and  $\mathcal{M}[\frac{p^+}{S}]$ ,  $\mathcal{M}[\frac{p^+}{S^2}]$  follow from the same residues (the latter with a double pole at  $p^+ = -k^+$ ). All six against quadrature at  $A = 0.62$ ,  $B = 1.45$ ,  $\kappa = 0.83$ ,  $\Lambda = 10^{-7}$ ,  $\vee = 40$ :

|         |                         |                   |
|---------|-------------------------|-------------------|
| h=1     | 0.28052425-0.62934122j  | agrees to 1.4e-15 |
| h=1/p   | 10.74337225-0.80137719j | agrees to 1.8e-15 |
| h=p     | 0.92920170+0.65250099j  | agrees to 2.3e-14 |
| h=1/S   | 0.32390943-0.33443150j  | agrees to 4.0e-16 |
| h=p/S   | -0.04338518-0.29490972j | agrees to 1.4e-15 |
| h=p/S^2 | 0.04623704-0.16604946j  | agrees to 4.2e-16 |

Only  $h = \frac{1}{p^+}$  is singular as  $\Lambda \rightarrow 0$ , and **what it produces is not**  $\log \Lambda$ :

$$\text{Ei}(-i\kappa\Lambda) \rightarrow \gamma_E + \log(|\kappa|\Lambda) - \frac{i\pi}{2} \text{sgn } \kappa = \gamma_E + \log \Lambda + \log \frac{|k \cdot (y' - z)|}{k^+} - \frac{i\pi}{2} \text{sgn } \kappa$$

(checked to six digits at  $\kappa = 0.83, 2.4$  and  $\Lambda = 10^{-6}, 10^{-8}$ ). **The rapidity logarithm arrives welded to a transverse logarithm.**

## 6. Can you use the + prescription? — No, not naively

At *fixed* transverse points there is nothing wrong with the identity. With  $g(p^+) = \frac{e^{-i\kappa p^+}}{p^+ A + k^+ B}$  and  $g(0) = \frac{1}{k^+ B}$ ,

$$\int_{\Lambda}^P \frac{dp^+}{p^+} g(p^+) = g(0) \log \frac{P}{\Lambda} + \int_0^P \frac{dp^+}{p^+} [g(p^+) - g(0)]$$

is exact. The damage is done by what happens next.

| $g(0)$ <b>deletes the <math>z</math>-phase.</b> The only thing damping the $z$ integral is $e^{-i\kappa p^+}$ with $\kappa = \frac{k \cdot (y' - z)}{k^+}$ ; setting $p^+ = 0$ removes it. Measured at large $ z $ : |                    |                    |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|
| $ z $                                                                                                                                                                                                                | subtraction $g(0)$ | full, $p^+ = 0.37$ | full, $p^+ = 0.01$ |
| $10^2$                                                                                                                                                                                                               | 0.212557193        | 1.26e-5            | 3.67e-3            |
| $10^3$                                                                                                                                                                                                               | 0.212517864        | 3.30e-8            | 6.96e-6            |
| $10^4$                                                                                                                                                                                                               | 0.212517471        | 1.12e-10           | 1.07e-8            |
| $10^5$                                                                                                                                                                                                               | <b>0.212517467</b> | 5.48e-13           | 4.61e-11           |
| The subtraction term sits on a flat plateau, equal to nine digits to                                                                                                                                                 |                    |                    |                    |
| $-\frac{1}{2} \frac{(x' - y') \cdot (x - w)}{(x' - y')^2 k^+ (x - w)^2} = -0.212517467,$                                                                                                                             |                    |                    |                    |
| so its $z$ integral is <b>logarithmically divergent</b> . The unsubtracted term at any $p^+ \neq 0$ falls away. <b>The prescription manufactures an infrared divergence that the row does not have.</b>              |                    |                    |                    |

The mechanism is that the damping scale depends on  $p^+$ :

$$|z|_{\max} \sim \frac{k^+}{|k| p^+} : \quad p^+ = 10^{-1} \rightarrow 12, \quad 10^{-2} \rightarrow 124, \quad 10^{-3} \rightarrow 1240, \quad 10^{-4} \rightarrow 12400,$$

visible in the table above as the  $p^+ = 0.01$  column decaying later than the  $p^+ = 0.37$  one. **The  $z$  and  $p^+$  integrations do not commute.**

### 6.1 What to do instead

1. **There is no UV step.** Nothing to regulate before the  $p^+$  integral.
2. **Do not split off  $\log \frac{\sqrt{-k^+}}{\Lambda}$  at fixed  $z$ .** Either keep the  $z$  and  $p^+$  integrals together, or introduce  $d = 2 - 2\epsilon$  on  $z$  *first* — not because the row needs it, but because it is what makes the subtraction term finite and the split legitimate.
3. **Then the prescription is safe, but the logarithm is not the usual one.** By §5 it comes as  $\log(|\kappa|\Lambda)$ , so the  $\log \frac{1}{\Lambda}$  multiplies a  $z$  integral that is itself log divergent, while its partner  $\log |\kappa| \sim \log |z|$  against  $\frac{d^2 z}{|z|^2}$  supplies a second logarithm. The row therefore produces a **mixed**  $\frac{1}{\epsilon} \log \frac{1}{\Lambda}$  **and a**  $\log^2$ , not a clean single rapidity logarithm riding on a leading-order structure. Plan the absorption into evolution accordingly.

zint/src/row14\_check.py (§1–2), row14\_div2.py (§3–4, §6), row14\_pplus.py, row14\_plus.py (§5–6), row14\_w.py (§4) — every number above.